# MILITARY STANDARD

**MIL-STD-1386-31**
**(NAVY)**

1 OCTOBER 1973

## LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

### ROCKET LAUNCHER, LAU-61/A, LAU-68/A OR LAU 69/A
### FLEET ISSUE UNIT LOAD

**UNIT LOAD DATA**

UNIT LOAD DRAWING ..... WR-54/174

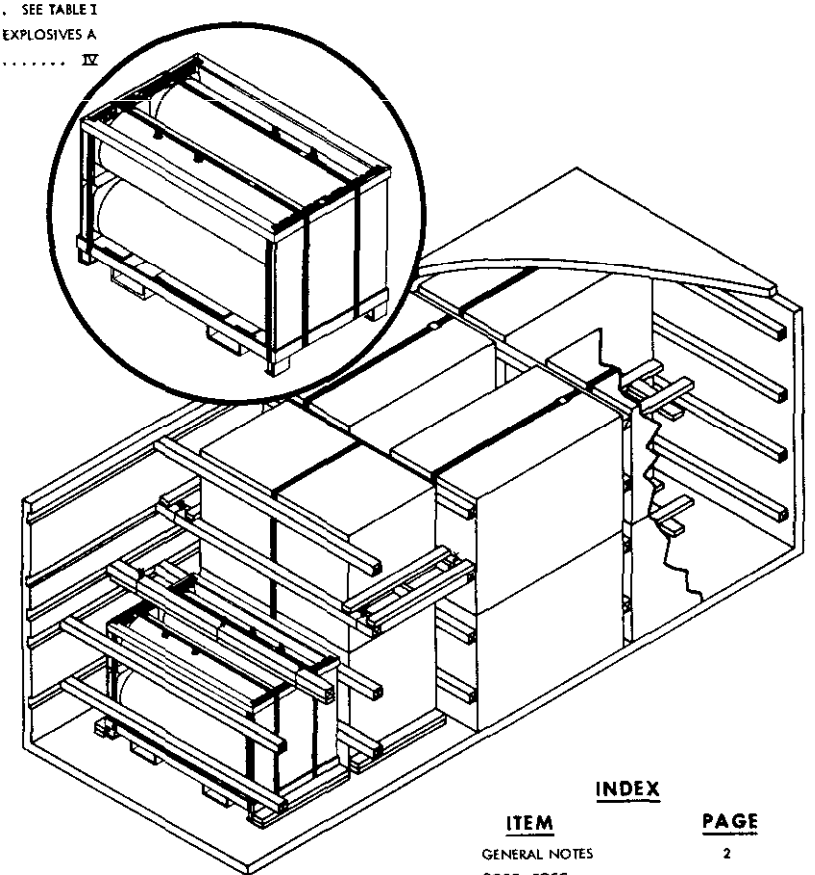
DIMENSIONS .. 61/A & 69/A - 62 3/8 L x 37 1/4 W x 41 7/8 H
.. 68/A 62 3/8 L x 37 1/4 W x 42 5/8 H

CUBE ..... 61/A & 69/A 56.1 CU. FT.
..... 68/A 57.3 CU. FT.

WEIGHT ..... SEE TABLE I

DOT HAZARD CLASSIFICATION ..... EXPLOSIVES A

COAST GUARD CLASS ..... IV



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| COFC, TOFC *CONTAINERSHIP* & HIGHWAY | 2 & 3 |
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NOTES:

1. *UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.*
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO THE INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

**AUTHORIZED FOR TRAILER-ON-FLAT-CAR (TOFC), CONTAINER-ON-FLAT-CAR (COFC), CONTAINERSHIP AND HIGHWAY SHIPMENTS**

**APPROVED BY US COAST GUARD**
R. M. Darnaby
DATE 31 Aug 1973

**APPROVED BY BUREAU OF EXPLOSIVES**
A. F. Grassmuck
DATE 9/4/73

M. S. Gray NWHL 9/14/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 9/14/73
SIGNATURE AIRSYSCOM BY DIRECTIONS DATE

**ORIGINATOR** A. R. Parinzi 9/14/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY



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# GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOADS OF ROCKET LAUNCHERS, LAU-61/A, LAU-68/A OR LAU-69A. THE UNIT LOADS SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/174.
2. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD- BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C.
3. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 83 INCH HEIGHTS FROM THE CONTAINER FLOOR.

**CAUTION**

WHEN POSITIONING CROSSMEMBERS, MAKE A VISUAL CHECK TO ENSURE THAT THE CROSSMEMBERS ARE LOCKED SECURELY IN PLACE.

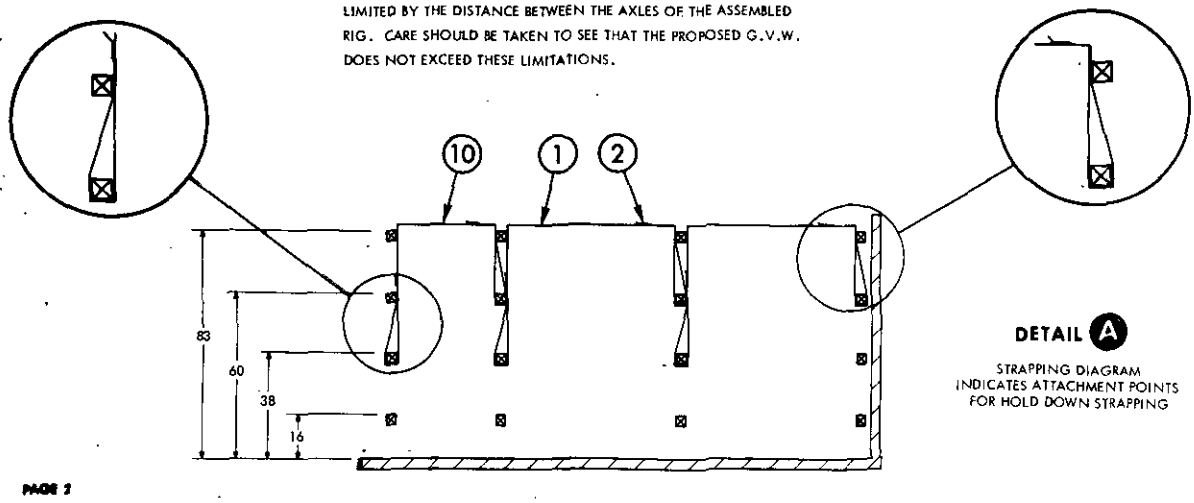
4. CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO BELT RAILS FOR SHIPMENT.
5. WHEN THE VOID BETWEEN THE CROSSMEMBER AND THE UNIT LOAD IS GREATER THAN ONE INCH, USE FILL MATERIAL TO SHIM BETWEEN CROSSMEMBER AND UNIT LOAD AS SHOWN IN DETAIL G.
6. SWAY BRACE, DETAIL F, IS ASSEMBLED IN PLACE. POSITION PIECE 6 IN UNIT LOAD FORK POCKETS. WITH PIECE 7 AGAINST UNIT LOAD, NAIL TO PIECE 6 WITH THREE 10d NAILS PER JOINT. CUT PIECE 8 TO FIT BETWEEN PIECES 7 AND NAIL TO PIECE 6 WITH THREE 10d NAILS.
7. SHIMS, PIECE NO. 11, SHALL BE OF SUFFICIENT THICKNESS TO FILL THE VOID BETWEEN THE TOP OF THE UNIT LOAD AND THE BOTTOM OF THE CROSSMEMBER. ONE HALF INCH SLACK IS PERMITTED.
8. LENGTH OF STRUT, PIECE 4, SHALL BE SUFFICIENT TO CAUSE DETAIL E TO FILL THE VOID BETWEEN THE UNIT LOAD AND THE CONTAINER SIDE WALL.
9. TWIST TIE BOTH ENDS OF DETAIL E TO CROSSMEMBER WITH 16 GAUGE TIE WIRE.
10. APPLICABLE MATERIAL SPECIFICATIONS:
    DUNNAGE LUMBER - FED. SPEC. MM-L-751
    NAILS - FED. SPEC. FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT.
    STRAP - FED. SPEC. QQ-S-781, TYPE I, CLASS A
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL CONTAINER DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 (VOL. 1).

# TOFC/COFC, CONTAINERSHIP AND HIGHWAY

1. THE CONTAINER LOAD CONSISTS OF 11 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS DOCUMENT.
2. THE LOADED MILVAN CONTAINER SHALL BE ON A 20 FT. MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES WHEN BEING MOVED IN TOFC SERVICE.
3. WHEN THE LOADED CONTAINER IS IN TOFC OR COFC SERVICE, IT SHALL BE ON FLAT RAILCARS THAT COMPLY WITH BUREAU OF EXPLOSIVES PAMPHLET 6C.
4. THE LOADED MILVAN MAY BE MOVED OVER PUBLIC HIGHWAYS ON A MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES OR ON SUITABLE COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY.

**NOTE**

IN CERTAIN STATES THE GROSS VEHICLE WEIGHT (G.V.W.) IS LIMITED BY THE DISTANCE BETWEEN THE AXLES OF THE ASSEMBLED RIG. CARE SHOULD BE TAKEN TO SEE THAT THE PROPOSED G.V.W. DOES NOT EXCEED THESE LIMITATIONS.




**DETAIL A**

STRAPPING DIAGRAM
INDICATES ATTACHMENT POINTS
FOR HOLD DOWN STRAPPING



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## CONTAINER LOAD DATA

NUMBER OF UNIT LOADS ..... 11
NUMBER OF CROSSMEMBERS REQD ..... 19
LOAD WEIGHT ..... SEE TABLE I
DUNNAGE WEIGHT ..... 215 POUNDS
CONTAINER WEIGHT ..... 5785 POUNDS
GROSS WEIGHT ..... SEE TABLE I

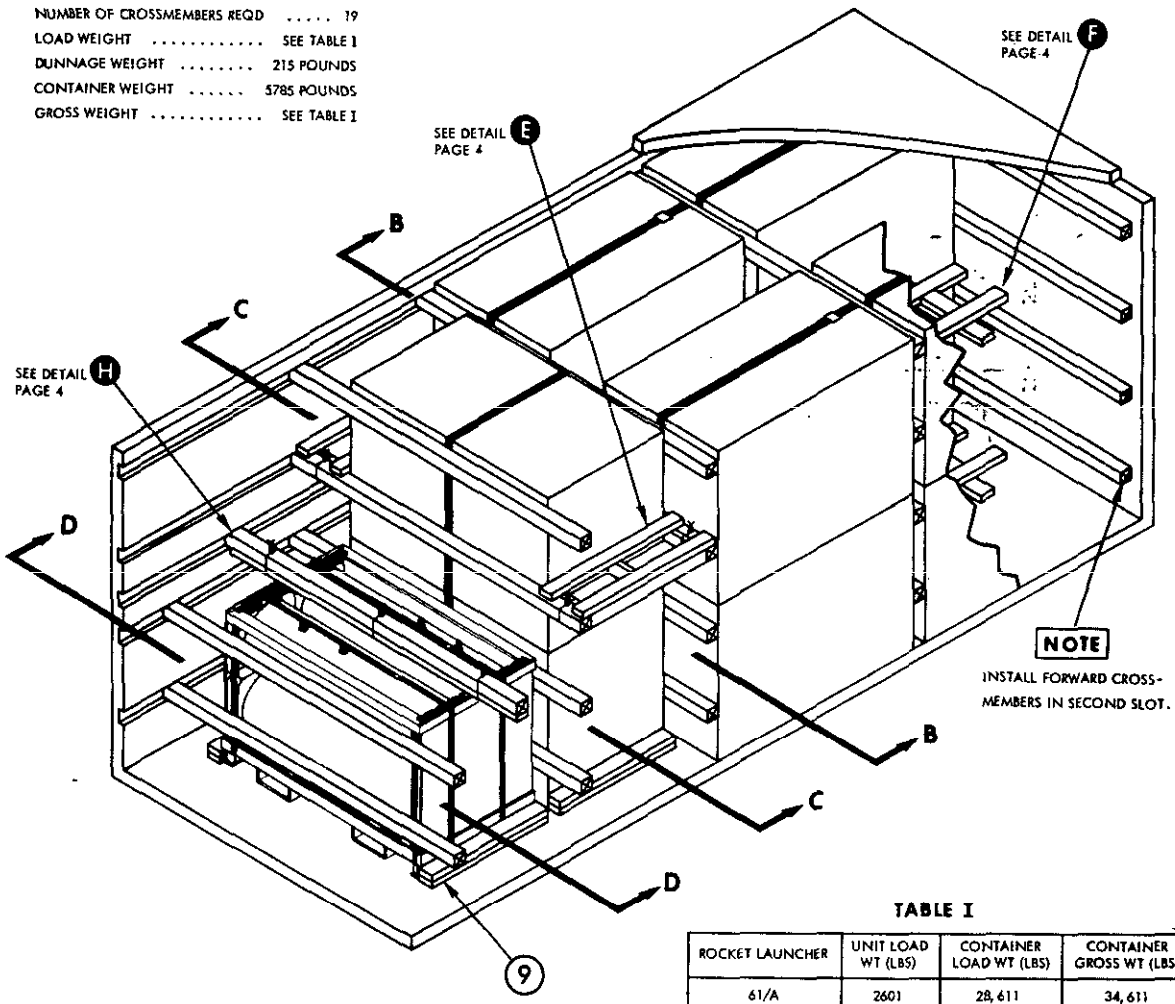
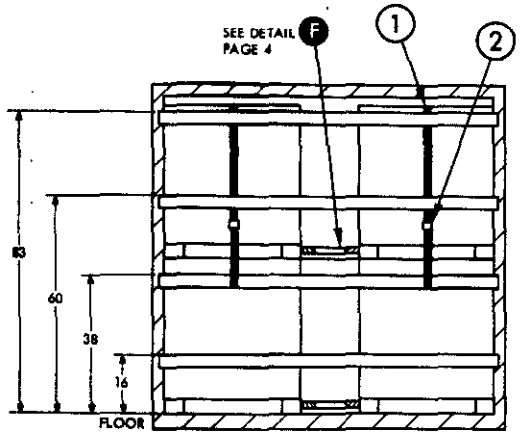



**TABLE I**

| ROCKET LAUNCHER | UNIT LOAD WT (LBS) | CONTAINER LOAD WT (LBS) | CONTAINER GROSS WT (LBS) |
|---|---|---|---|
| 61/A | 2601 | 28,611 | 34,611 |
| 69/A | 2457 | 27,027 | 33,027 |
| 68/A | 2436 | 26,796 | 32,796 |



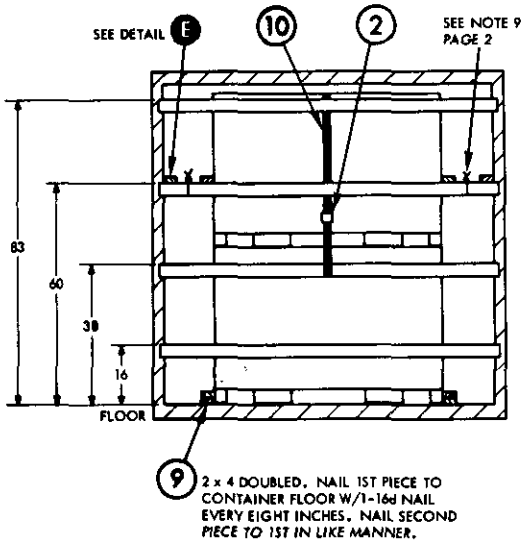

**SECTION B-B**

| PIECE NO. | DESCRIPTION | SIZE | NO. PCS REQD | NAIL TO | NUMBER | SIZE |
|---|---|---|---|---|---|---|
| | | | | | NAILS | |
| 11 | SHIM | 4 WIDE x 90 LONG x THICKNESS TO SUIT | AS REQD | SEE GENERAL NOTE 7 | | |
| 10 | STRAP | 1 1/4 x .035 x 11 FT | 1 | - | - | - |
| 9 | SLEEPER | 2 x 4 x 37 | 8 | SEE SECTIONS C&D | - | - |
| 8 | SPACER | 2 x 4 x TO SUIT | 8 | 6 | 3 | 10d |
| 7 | STRINGER | 2 x 4 x 60 | 8 | 6 | 3 PER JOINT | 10d |
| 6 | SUPPORT | 2 x 4 x 29 | 8 | SEE 7 | - | - |
| 5 | SUPPORT | 2 x 4 x 42 1/2 | 4 | 4<br>3 | 1 PER JOINT<br>5 | 12d<br>12d |
| 4 | STRUT | 2 x 4 x TO SUIT | 6 | SEE 3 | - | - |
| 3 | LONGITUDINAL | 2 x 4 x 37 | 4 | 4 | 2 PER JOINT | 16d |
| 2 | SEAL | 1 1/4 | 10 | - | - | - |
| 1 | STRAP | 1 1/4 x .035 x 14 FT | 4 | - | - | - |

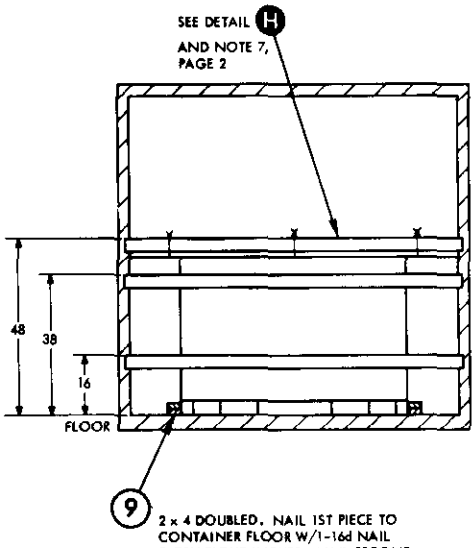
LIST OF MATERIALS AND NAILING DATA



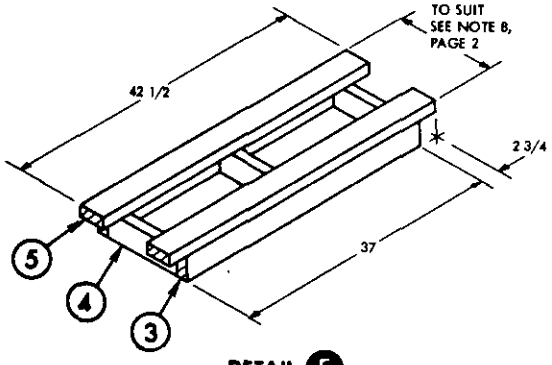
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**SECTION C-C**

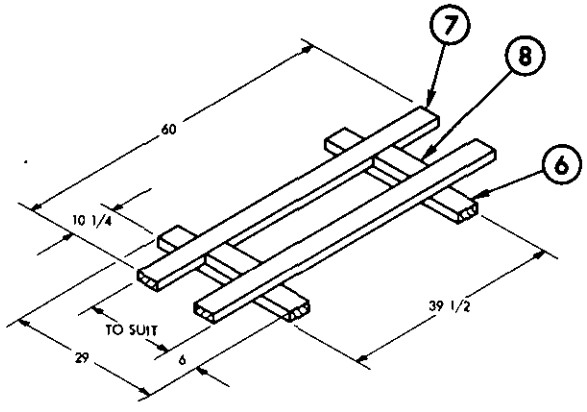



**SECTION D-D**



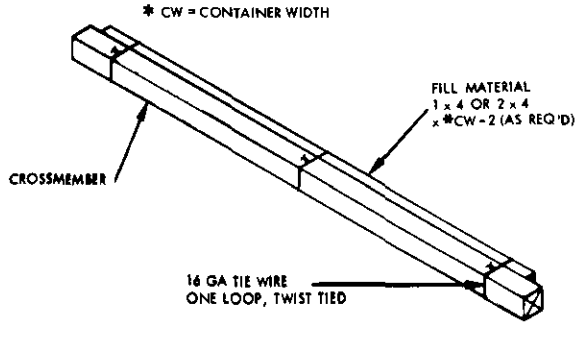

**DETAIL E**

SWAY BRACE FRAME
2 REQUIRED
SEE NOTE 9, PAGE 2



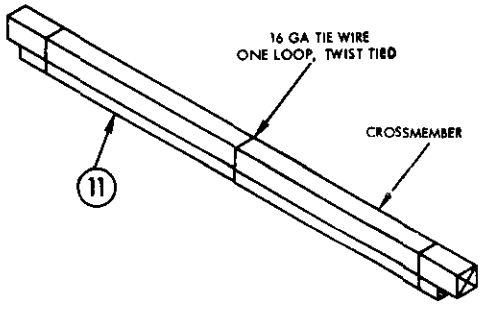

**DETAIL F**

SWAY BRACE FRAME
4 REQUIRED
TO BE ASSEMBLED IN PLACE
SEE NOTE 6




**DETAIL G**

FILL MATERIAL
SEE GENERAL NOTE 5




**DETAIL H**

HOLD DOWN SHIM ASSEMBLY
SEE GENERAL NOTE 7



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REVIEW ACTIVITIES:

NAVY - OS, AS

PREPARING ACTIVITY:

NAVY - OS
(PROJECT NO. 8140-N207)



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